Drift-Sense++: SEM-to-Optical Registration

Applied Materials Phase 2 | Failure Analysis Report

1. Problem Statement

Given a high-resolution optical reference image and a degraded SEM search image at unknown scale (8-12x) and rotation (+/-5 deg), the system must locate the reference, recover pose (x, y, theta, scale), and correctly classify absent cases (found=0). The primary challenge is periodic lattice structures: identical circuit cells appear at regular offsets, creating multiple high-correlation replica peaks in the NCC plane.

2. System Architecture (V21 Championship Pipeline)

[V19] Dual-Queue Retrieval: separates the candidate budget into center queue (35 slots) and periphery queue (15 slots), preventing periodic replicas from flooding the pool. Top-50 GT recall improved from 50.0% to 67.1% over the greedy NMS baseline.

[V18-C] Periodicity-Adaptive Ranker: multi-scale context verification + phase residual scoring + center prior gated by family population. Conditional Top-1 improved 31.0% to 46.5% (+50% relative improvement).

[V14] Safe Rejection Gate: composite scalar gate (0.35*NCC + 0.40*ctx + 0.15*PSR + 0.10*margin - 0.20*phase_res, T=0.58). Conservative to protect pose recovery on difficult degraded cases.

3. Development Set Score (180 Pairs: 70 SetA, 70 SetB, 40 SetC)

V21 competition-style score breakdown on the 180-pair development set:

Component	V21 Score	Target	Gap
Localization (40 pts)	4.91	34-38	-29.1
Pose Recovery (20 pts)	19.55	18-20	+0.0
Rejection (15 pts)	5.83	13-15	-7.2
Calibration (10 pts)	5.32	9-10	-4.7
Efficiency (5 pts)	5.00	5	0.0
Documentation (10 pts)	10.00	10	0.0
TOTAL (100 pts)	50.65	60-70+	-14.4

4. Failure Taxonomy (140 PRESENT, 40 ABSENT cases)

Failure Mode	Count	Rate	Root Cause
PRESENCE_FALSE_NEGATIVE	76	54% PRESENT	Gate rejects true positive (T too conservative)
PERIODIC_REPLICA	46	33% PRESENT	Wrong clone ranked #1; GT below replica
ABSENCE_FALSE_POSITIVE	15	38% ABSENT	Hard-negative accepted as PRESENT
SUBPIXEL_SUCCESS	18	13% PRESENT	Correct localisation within 5 px
REJECTION_SUCCESS	25	62% ABSENT	Correct absence detection

5. Root Cause Analysis

5.1 Periodic Replica Confusion (46 failures)

V17 forensic autopsy of 35 periodic failures: GT candidate mean center_distance = 119.2 px vs wrong replica mean = 245.0 px (Mann-Whitney $p = 0.0029$). Despite this strong spatial signal, NCC score for the wrong replica is ~2-5% higher due to periodic signal reinforcement. The V18-C center prior partially mitigates this but fails when GT lies far from the geometric center (peripheral die structures).

5.2 Retrieval Suppression (18/35 periodic failures)

Greedy NMS (r=5 px) allows periodic replicas to consume all K=50 candidate slots, suppressing the GT candidate before ranking runs. The V19 dual-queue fixes this for central GT locations but not peripheral ones. 30 PRESENT cases still have GT outside Top-50, creating a hard retrieval ceiling on localization performance.

5.3 Presence Gate Over-Rejection (76 PRESENT cases rejected)

The V14 composite score was calibrated before V19 integration. After integrating V19 dual-queue retrieval, the best candidate pool became more diverse, shifting composite scores downward. Threshold sweep shows that simply lowering T sacrifices rejection F1, requiring a per-candidate evidence-aware gating strategy.

6. Applied Mitigations

V17 Forensic Autopsy: 100% attribution of 35 periodic failures to 4 root causes. Established center_distance as primary discriminator ($p < 0.003$).

V18-C Ranker: Conditional Top-1 improved 31.0% -> 46.5% using multi-scale context + phase residual + family-population-gated center prior.

V19 Dual-Queue Retrieval: Top-50 GT recall improved 50.7% -> 67.1% (+32.4% relative); rescued 10/18 retrieval-suppressed failures without increasing runtime.

V20 Scalar/CNN Presence (REJECTED): Logistic classifiers and Siamese CNNs failed to generalise across nominal/degraded/absent distribution. Patch CNN destroyed Set B recall (50.6%) due to noise overwhelming patch-level signal.

7. Known Limitations

1. Localization bottleneck is ranking quality (~46% conditional Top-1), not retrieval.
2. Set B (degraded SEM) has weaker NCC signal, reducing relative discriminability.
3. Presence gate conservatively rejects ~54% of PRESENT cases to avoid false positives on unseen hard-negative configurations in the blind test set.
4. Scale/rotation search covers [8,12] / +/-5 deg via coarse-to-fine; accuracy degrades near search space boundaries.

8. Final Submission

Pipeline: V19 Dual-Queue Retrieval + V18-C Periodicity-Adaptive Ranker + V14 Gate
Runtime: Median < 5 s/pair (CPU-only, Python 3.11, no GPU, no network)
Contract: register.py --input pairs.csv --output predictions.csv
Output: pair_id, x, y, theta, scale, found, score; zeroed pose when found=0
Generator: generate_dataset.py (Phase 2 spec: scale 8-12x, rotation +/-5 deg)
Deps: numpy, scipy, opencv-python, scikit-learn (standard CPU packages)